IMO 2026 — Problem 5

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-05/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $\mathbb{R}_{>0}$ be the set of positive real numbers. Determine all functions $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ such that $\sqrt{\frac{x^2+f(y)^2}{2}} \geq \frac{f(x)+y}{2} \geq \sqrt{xf(y)}$ for every $x, y \in \mathbb{R}_{>0}$.

Solution document

Status

solved

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Approaches tried

- **Iteration and rigidity (successful):** Substituting $x = f(y)$ forces an affine relation along every forward orbit. A symmetrized substitution then gives a quadratic modulus estimate for the displacement $g = f - \mathrm{id}$, from which monotonicity, continuity, and finally constancy follow.
- **Direct inequality reformulation (exploratory):** Rewriting both inequalities in terms of $g(x) - g(y)$ reveals error terms vanishing when $x = f(y)$, but the raw asymmetric bounds do not by themselves immediately compare all displacement classes. The iterate identity supplies the needed symmetry.
- **Sanity checking:** Random numerical tests verify both inequalities for translations $f(x) = x + c$ over many orders of magnitude; this is only a check, not part of the proof.

Current best

The complete set of solutions is

$$\boxed{f(x) = x + c \quad \text{for all } x > 0, \qquad c \geq 0.}$$

The proof first derives $f(f(t)) = 2f(t) - t$. For $g(t) = f(t) - t$, forward iteration gives $f^n(t) = t + ng(t)$ and hence $g \geq 0$. Comparing the original inequality at $(x, y) = (f(u), v)$ and $(f(v), u)$ gives a quadratic estimate on $|g(u) - g(v)|$. This forces f to be increasing, then g to be nondecreasing and continuous, and a partition argument forces g to be constant.

Full proof

Define

$$g(t) = f(t) - t \qquad (t > 0).$$

1. The iterate identity

In the given chain, set $x = f(y)$. Its two outer terms are both $f(y)$, since

$$\sqrt{\frac{f(y)^2 + f(y)^2}{2}} = f(y) \quad \text{and} \quad \sqrt{f(y)f(y)} = f(y).$$

Thus equality holds throughout, and

$$\frac{f(f(y)) + y}{2} = f(y).$$

Therefore

$$f(f(y)) = 2f(y) - y. \qquad (1)$$

In terms of g, equation (1) says

$$g(f(y)) = f(f(y)) - f(y) = f(y) - y = g(y). \qquad (2)$$

It follows by induction from (2) that, for every integer $n \geq 0$,

$$f^n(y) = y + ng(y). \qquad (3)$$

Every forward iterate belongs to $\mathbb{R}_{>0}$. If $g(y) < 0$, the right side of (3) is nonpositive for all sufficiently large n, a contradiction. Hence

$$g(y) \geq 0 \qquad \text{for all } y > 0. \qquad (4)$$

2. A quadratic comparison estimate

Fix $u, v > 0$. Apply the original chain with $x = f(u)$ and $y = v$. By (1), and by $v = f(v) - g(v)$, its middle numerator is

$$f(f(u)) + v = f(u) + g(u) + f(v) - g(v).$$

Writing $X = f(u)$, $Z = f(v)$, and $d = g(u) - g(v)$, we obtain

$$\sqrt{\frac{X^2 + Z^2}{2}} \geq \frac{X + Z + d}{2} \geq \sqrt{XZ}. \qquad (5)$$

The right-hand inequality in (5) gives

$$d \ge 2\sqrt{XZ} - X - Z = -(\sqrt{X} - \sqrt{Z})^2. \qquad (6)$$

Interchanging u and v in the same argument replaces d by $-d$, so (6) also gives

$$-d \ge -(\sqrt{X} - \sqrt{Z})^2.$$

Consequently

$$|g(u) - g(v)| \le \left(\sqrt{f(u)} - \sqrt{f(v)}\right)^2. \qquad (7)$$

3. Monotonicity

We first prove that f is strictly increasing. Let $u > v$, and set

$$h = u - v > 0, \qquad d = g(u) - g(v), \qquad \Delta = f(u) - f(v) = h + d.$$

If $\Delta = 0$, (7) gives $d = 0$, contrary to $\Delta = h + d$ and $h > 0$.

Suppose instead that $\Delta < 0$. Then $d < -h$, while (7) yields

$$|d| \le \left(\sqrt{f(u)} - \sqrt{f(v)}\right)^2 = |\Delta| \frac{|\sqrt{f(u)} - \sqrt{f(v)}|}{\sqrt{f(u)} + \sqrt{f(v)}} < |\Delta|.$$

On the other hand, since $\Delta = h + d < 0$,

$$|\Delta| = |d| - h < |d|,$$

a contradiction. Thus $\Delta > 0$, proving that f is strictly increasing.

Every iterate f^n is consequently strictly increasing. From (3), for $u > v$ and every positive integer n,

$$u + ng(u) = f^n(u) > f^n(v) = v + ng(v).$$

Hence

$$g(u) - g(v) > -\frac{u - v}{n}.$$

Letting $n \to \infty$ gives $g(u) \ge g(v)$. Therefore g is nondecreasing. (8)

4. Continuity

Fix $t > 0$. Since g is nondecreasing, its right limit

$$L = \lim_{s \downarrow t} g(s)$$

exists and is finite (for example, $g(t) \le g(s) \le g(t+1)$ when $t < s < t + 1$). Put $J = L - g(t) \ge 0$ and $A = f(t) > 0$. As $s \downarrow t$, we have

$$f(s) = s + g(s) \longrightarrow t + L = A + J.$$

Passing to the limit in (7) gives

$$J \le (\sqrt{A + J} - \sqrt{A})^2. \qquad (9)$$

If $J > 0$, then

$$(\sqrt{A+J} - \sqrt{A})^2 = J\frac{\sqrt{A+J} - \sqrt{A}}{\sqrt{A+J} + \sqrt{A}} < J,$$

contradicting (9). Hence $J = 0$, so g is right-continuous at t.

The left limit exists as well (when approaching within $\mathbb{R}_{>0}$). If

$$J = g(t) - \lim_{s \uparrow t} g(s) > 0,$$

then passing to the limit in (7) gives

$$J \le (\sqrt{f(t)} - \sqrt{f(t) - J})^2 < J,$$

again a contradiction. Thus g is also left-continuous. Therefore g, and hence $f(t) = t + g(t)$, is continuous on $\mathbb{R}_{>0}$. (10)

5. The displacement is constant

Fix $0 < a < b$. For a positive integer N, partition $[a, b]$ into N equal subintervals, with endpoints

$$x_i = a + \frac{i(b-a)}{N} \qquad (0 \le i \le N).$$

Because f is strictly increasing and g is nondecreasing, the quantities

$$\Delta_i = f(x_i) - f(x_{i-1}) > 0, \qquad d_i = g(x_i) - g(x_{i-1}) \ge 0$$

are nonnegative as indicated. By (7),

$$\begin{aligned} d_i &\le \left(\sqrt{f(x_i)} - \sqrt{f(x_{i-1})}\right)^2 \\ &= \frac{\Delta_i^2}{\left(\sqrt{f(x_i)} + \sqrt{f(x_{i-1})}\right)^2} \le \frac{\Delta_i^2}{4f(a)}. \end{aligned}$$

Summing and using telescoping gives

$$\begin{aligned} 0 \le g(b) - g(a) &\le \frac{1}{4f(a)} \sum_{i=1}^{N} \Delta_i^2 \\ &\le \frac{\max_i \Delta_i}{4f(a)} \sum_{i=1}^{N} \Delta_i \\ &= \frac{\max_i \Delta_i}{4f(a)} (f(b) - f(a)). \qquad (11) \end{aligned}$$

By (10), f is uniformly continuous on the compact interval $[a, b]$. The partition mesh tends to zero, so $\max_i \Delta_i \to 0$ as $N \to \infty$. Equation (11) therefore gives $g(b) = g(a)$. Since $a < b$ were arbitrary, g is constant on $\mathbb{R}_{>0}$.

Thus there is a constant c such that

$$f(x) = x + c.$$

By (4), $c \ge 0$.

6. Verification

Conversely, let $c \geq 0$ and define $f(x) = x + c$. Then f maps positive reals to positive reals, and

$$\frac{f(x) + y}{2} = \frac{x + (y + c)}{2} = \frac{x + f(y)}{2}.$$

For the two positive numbers x and $f(y)$, the root-mean-square/arithmetic-mean inequality and the arithmetic-mean/geometric-mean inequality give

$$\sqrt{\frac{x^2 + f(y)^2}{2}} \geq \frac{x + f(y)}{2} \geq \sqrt{xf(y)}.$$

Hence every such function works, completing the characterization.